```
JupyterLab and choose **Run All**.\n",
   "\n",
   "The dataset contains 891 passenger records
and 12 columns. `Survived` is the target
variable (0 = did not survive, 1 = survived). This
is an exploratory analysis, not a causal study.\n"
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   "import numpy as np\n",
   "import matplotlib.pyplot as plt\n",
   "import seaborn as sns\n",
   "\n",
   "sns.set_theme(style=\"whitegrid\")\n",
   "df = pd.read_csv(\"titanic_dataset.csv\")\n",
   "\n",
   "print(\"Shape:\", df.shape)\n",
   "display(df.head())\n"
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  "df.info()\n"
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  "display(df.describe(include=\"all\").T)\n"
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  "## 3. Missing values and duplicates"
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  "missing =
df.isnull().sum().sort_values(ascending=False)
\n",
  "display(missing[missing >
0].to_frame(\"Missing values\"))\n",
  "print(\"Duplicate rows:\",
df.duplicated().sum())\n"
 ]
```

```
df.duplicated().sum())\n"
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    "for col in [\"Survived\", \"Pclass\", \"Sex\",
\"Embarked\", \"SibSp\", \"Parch\"]:\n",
    "    print(f\"\\\\nValue counts for {col}:\")\n",
    "
display(df[col].value_counts(dropna=False))\n"
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  "# Create a readable survival label for
charts\n",
  "df[\"SurvivalStatus\"] =
df[\"Survived\"].map({0: \"Did not survive\", 1:
\"Survived\"})\n",
  "print(\"Overall survival rate: {:.1f}%
\".format(df[\"Survived\"].mean() * 100))\n"
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  "## 6. Univariate analysis: histograms and
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\n",
   "sns.histplot(data=df, x=\"Age\", bins=25,
kde=True, ax=axes[0])\n",
   "axes[0].set_title(\"Age distribution\")\n",
   "sns.countplot(data=df, x=\"SurvivalStatus\",
ax=axes[1])\n",
   "axes[1].set_title(\"Passenger survival
counts\")\n",
   "plt.tight_layout()\n",
   "plt.show()\n"
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   "**Observation:** Age is concentrated
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fewer very young and older passengers. More
passengers did not survive than survived."
  ]
```

```
 ]
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   "fig, axes = plt.subplots(1, 2, figsize=(12, 4))
\n",
   "sns.barplot(data=df, x=\"Sex\",
y=\"Survived\", errorbar=None, ax=axes[0])\n",
   "axes[0].set_title(\"Survival rate by sex\")\n",
   "axes[0].set_ylabel(\"Survival rate\")\n",
   "sns.barplot(data=df, x=\"Pclass\",
y=\"Survived\", errorbar=None, ax=axes[1])\n",
   "axes[1].set_title(\"Survival rate by passenger
class\")\n",
   "axes[1].set_ylabel(\"Survival rate\")\n",
   "plt.tight_layout()\n",
```

```
   "plt.tight_layout()\n",
   "plt.show()\n",
   "\n",
   "display((df.groupby(\"Sex\")
[\"Survived\"].mean() *
100).round(1).rename(\"Survival rate (%)\"))\n",
   "display((df.groupby(\"Pclass\")
[\"Survived\"].mean() *
100).round(1).rename(\"Survival rate (%)\"))\n"
  ]
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   "**Observation:** Female passengers had a
substantially higher survival rate than male
passengers. First-class passengers had a
higher survival rate than second- and
third-class passengers. These are associations
in this dataset."
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```
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  "plt.figure(figsize=(8, 4))\n",
  "sns.boxplot(data=df, x=\"SurvivalStatus\",
y=\"Fare\", showfliers=False)\n",
  "plt.title(\"Fare distribution by survival
status\")\n",
  "plt.xlabel(\"\")\n",
  "plt.show()\n"
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  "**Observation:** The fare distribution is
right-skewed. Survivors generally show a higher
median fare, consistent with the class-related
survival pattern. High fares are potential
outliers and should not be removed without
justification."
 ]
```

```
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  "sns.scatterplot(data=df, x=\"Age\",
y=\"Fare\", hue=\"SurvivalStatus\",
alpha=0.65)\n",
  "plt.title(\"Age versus fare, coloured by
survival\")\n",
  "plt.show()\n"
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   "pair_data = df[[\"Survived\", \"Pclass\", \"Age\", \"SibSp\", \"Parch\", \"Fare\"]].dropna()\n",
   "sns.pairplot(pair_data, hue=\"Survived\", corner=True, diag_kind=\"hist\")\n",
   "plt.show()\n"
  ]
```

```
  "plt.show()\n"
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  "**Observation:** Pairwise plots help
compare numeric variables and survival
groups. Pclass and Fare show an inverse
association because lower class numbers
represent higher passenger classes and
generally higher fares."
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  "## 11. Correlation heatmap"
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   "numeric = df[[\"Survived\", \"Pclass\",
\"Age\", \"SibSp\", \"Parch\", \"Fare\"]]\n",
   "corr = numeric.corr(numeric_only=True)\n",
   "plt.figure(figsize=(8, 6))\n",
   "sns.heatmap(corr, annot=True,
cmap=\"coolwarm\", center=0, fmt=\".2f\")\n",
   "plt.title(\"Correlation heatmap\")\n",
   "plt.show()\n"
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   "**Observation:** Survived has a negative
correlation with Pclass (higher class number,
lower survival) and a positive correlation with
Fare. Correlation does not establish cause and
may reflect related factors."
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   "## 12. Key findings and conclusion"
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  "source": [
   "## 12. Key findings and conclusion"
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   "- The dataset contains **891 passengers**
and **12 original columns**.\n",
   "- Overall survival was approximately
**38.4%**; the majority did not survive.\n",
   "- Survival differed strongly by sex: women
had a much higher survival rate than men.\n",
   "- First-class passengers had the highest
survival rate, followed by second and third
class.\n",
   "- Age and Fare contain missing values; Cabin
is missing for most passengers. Missing values
should be handled deliberately before
predictive modelling.\n",
   "- Fare is right-skewed, with a small number
of high-fare observations.\n",
   "- The analysis is descriptive. Observed
relationships do not prove causation.\n",
   "\n",
   "**Conclusion:** Sex, passenger class and
fare are visibly associated with survival in this
sample. Further analysis could explore
```

```
   "- Fare is right-skewed, with a small number of high-fare observations.\n",
   "- The analysis is descriptive. Observed relationships do not prove causation.\n",
   "\n",
   "**Conclusion:** Sex, passenger class and fare are visibly associated with survival in this sample. Further analysis could explore interactions among age, family size, embarkation port and passenger class. The notebook preserves raw records and does not silently delete outliers or fill missing values."
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